

NETWORKED CONTROL

23/01/2023

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EXERCIZES

1) Consider the following continuous-time model of system $\mathcal{S}$.

$$\dot{x}(t) = x(t) + 2u(t)$$

$$y(t) = x(t)$$

A. Answer to the following questions.

- Compute the eigenvalue and the mode of the model. Is the system open-loop asymptotically stable?
- Assume that the system is controlled using a continuous-time control law of the type $u(t) = Ky(t)$. Define the range of values of K that make the closed-loop system asymptotically stable.

B. Compute the parameters F, G , and H of the discrete-time dynamical model

$$x_{k+1} = Fx_k + Gu_k, \quad y_k = Hx_k$$

of system $\mathcal{S}$ with sampling time $h = \log_e 2$ s.

Also, assuming that the system is controlled using a digital control law of the type $u_k = Ky_k$, define the range of values of K that make the closed-loop system asymptotically stable.

C. Assume that the system $\mathcal{S}$ is controlled through the networked control system (NCS) sketched in the following figure.

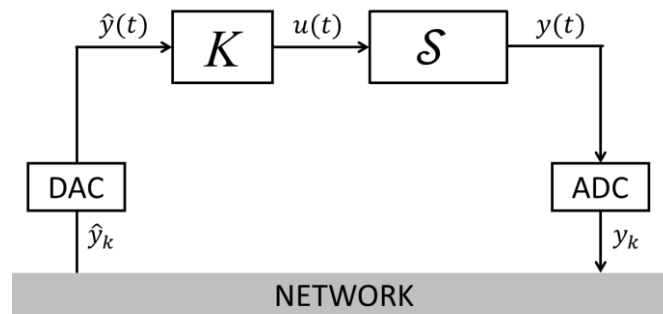


Figure 1: Networked control system

- Assume that $h = \log_e 2$ s and $K = -1$ and that there are no dropouts. What is the maximum value $\bar{\tau}$ of the delay τ such that the NCS is asymptotically stable iff $\tau < \bar{\tau}$?
- Assume now that the NCS is subject to packet dropouts, that the delay is absent (i.e., $\tau = 0$ s), and that $h = \log_e 2$. Variable θ_k has the following periodic behavior.

k	1	2	3	4	5	6	7	8	9	...
θ_k	1	0	1	0	1	0	1	0	1	...

Define the range of values of K that make the NCS asymptotically stable.

2) Consider the NCS displayed in Figure 1 and assume that (i) the sampling time h_k is constant, i.e., $h_k = h$ for all k ; (ii) the network is subject to constant delay τ . Answer to the following questions.

- A. Define the model of the NCS subject to dropouts.
- B. Define the asymptotic dropout rate $\bar{r}$.
- C. Provide the definition of the *maximal dropout rate* and discuss (in a concise but complete way) on how it is possible to compute $\bar{r}$.
- D. How is it possible to exploit the knowledge of $\bar{r}$ to design a NCS with reduced network load?

3) Given a linear time invariant system

$$\dot{x}(t) = Ax(t) + Bu(t)$$

where $u(t) \in \mathbb{R}$ is a scalar input. Answer to the following questions.

- Define the **stabilizability** of the system.
- In case we have an **admissible discrete control set** $\mathcal{U} = \{\bar{u}_1, \dots, \bar{u}_N\}$, define the **boundability** of the system.
- Provide a comparison between **boundability** and **stabilizability**.
- Consider the system

$$\begin{cases} \dot{x}_1(t) = 2x_1(t) - x_2(t) + u(t) \\ \dot{x}_2(t) = -x_1(t) + 2x_2(t) \end{cases}$$

Is it stabilizable? In case a digital quantized controller is used to control it with sampling time $h = 0.2$ s, what is the minimum number of bits N_B encoding the control input required to make the system boundable?